1.

Ans- The derivative of a function measures its rate of change with respect to x.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Basic derivative rules:

1. Power Rule: $\frac{d}{dx} x^n = nx^{n-1}$

2. Constant Rule: $\frac{d}{dx}(c) = 0$

3. Sum/difference Rule: $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) + g'(x)$

2.

Ans- The chain Rule is used to differentiate composite functions by differentiating outer function first and multiplying derivative of inner function

Formula: $\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$

Given: $y = (3x^2 + 5)^4$

Applying Chain Rule: $\frac{dy}{dx} = 4(3x^2 + 5)^3 \times 6x$

$$\frac{dy}{dx} = 24x(3x^2 + 5)^3$$

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Q3 Differentiability, Continuity and mean Value theorem.

Ans

① Relationship between differentiability and Continuity.

- If a function is differentiable at an point it is Continus there
- A ~~Continue~~ Continious function may not always be differentiable

② Mean Value theorem (MVT)

If a function $f(x)$ satisfies

① It is Continuous on $[a, b]$

② It is differentiable on (a, b).

then there exist at least one point c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$